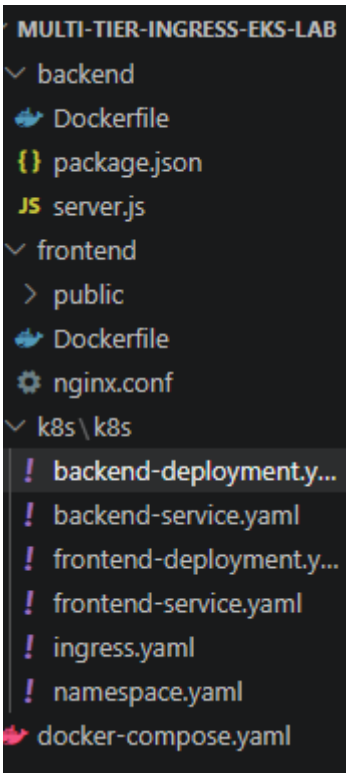
Git clone:

```
joaoribeiro@JOCAPC:~$ git clone https://github.com/professordiogodev/multi-tier-ingress-eks-
lab
Cloning into 'multi-tier-ingress-eks-lab'...
remote: Enumerating objects: 14, done.
remote: Counting objects: 100% (14/14), done.
remote: Compressing objects: 100% (14/14), done.
remote: Total 14 (delta 0), reused 14 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (14/14), done.
joaoribeiro@JOCAPC:~$ cd multi-tier-ingress-eks-lab
joaoribeiro@JOCAPC:~/multi-tier-ingress-eks-lab$ tree
.
├── backend
│   ├── Dockerfile
│   ├── package.json
│   └── server.js
├── docker-compose.yaml
└── frontend
    ├── Dockerfile
    ├── nginx.conf
    └── public
        ├── app.js
        ├── index.html
        └── style.css

4 directories, 9 files
```

Multi-tier architecture:



Backend Deployment:

k8s > k8s > ! backend-deployment.yaml > {} spec > {} template > {} spec > [] containe

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: backend
  labels:
    app: backend
spec:
  replicas: 2
  selector:
    matchLabels:
      app: backend
  template:
    metadata:
      labels:
        app: backend
    spec:
      containers:
        - name: backend
          image: joaoribeiro9595/lab-backend:1.0
          ports:
            - containerPort: 3000
          readinessProbe:
            httpGet:
              path: /readyz
              port: 3000
          livenessProbe:
            httpGet:
              path: /healthz
              port: 3000
```

Frontend Deployment:

k8s > k8s > ! frontend-deployment.yaml > {} spec > {} template > {} spec > [] cont

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend
  labels:
    app: frontend
spec:
  replicas: 2
  selector:
    matchLabels:
      app: frontend
  template:
    metadata:
      labels:
        app: frontend
    spec:
      containers:
        - name: frontend
          image: joaoribeiro9595/lab-frontend:1.0
          ports:
            - containerPort: 80
          readinessProbe:
            httpGet:
              path: /
              port: 80
          livenessProbe:
            httpGet:
              path: /
              port: 80
```

backend service:

k8s > k8s > ! backend-service.yaml > {} spec >

```yaml
apiVersion: v1
kind: Service
metadata:
  name: backend
spec:
  type: ClusterIP
  selector:
    app: backend
  ports:
    - port: 3000
      targetPort: 3000
```

frontend service:

k8s > k8s > ! frontend-service.yaml > {} spec

```yaml
apiVersion: v1
kind: Service
metadata:
  name: frontend
spec:
  type: ClusterIP
  selector:
    app: frontend
  ports:
    - port: 80
      targetPort: 80
```

ingress rules:

```yaml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: app-ingress
spec:
  ingressClassName: nginx
  rules:
    - host: app.joao.diogohack.shop
      http:
        paths:
          - path: /api
            pathType: Prefix
            backend:
              service:
                name: backend
                port:
                  number: 3000
          - path: /
            pathType: Prefix
            backend:
              service:
                name: frontend
                port:
                  number: 80
```

k8s namespace:

```
k8s > k8s > ! namespace.yaml > {} metadata
1  apiVersion: v1
2  kind: Namespace
3  metadata:
4    name: demo-joao
5
```

Testing the browser via LB with a Host header:

```
joaoribeiro@JOCAPC:~/multi-tier-ingress-eks-lab$ curl -H "Host:$HOST" http://adca49bf6fa4542
a5a129dbe476c5321-e63121efeb8e4144.elb.us-east-1.amazonaws.com/api/ping
{"ok":true,"message":"pong","service":"backend","time":"2026-05-06T12:59:29.140Z","hostname"
:"backend-59947449d7-cnf2r"}joaoribeiro@JOCAPC:~/multi-tier-ingress-eks-lab$
```

Route 53 DNS:

Pods from my demo-joao:

```
joaoribeiro@JOCAPC:~/multi-tier-ingress-eks-lab$ kubectl get pods -n demo-joao -o wide
NAME                        READY   STATUS    RESTARTS   AGE    IP              NODE                                       NOMINATED NODE   READINESS GATES
backend-59947449d7-4xpmm    1/1     Running   0          132m   192.168.2.132   ip-192-168-20-198.ec2.internal   <none>           <none>
backend-59947449d7-cnf2r    1/1     Running   0          132m   192.168.36.63   ip-192-168-57-231.ec2.internal   <none>           <none>
frontend-79ccb5bc9f-jbbk6   1/1     Running   0          131m   192.168.31.107  ip-192-168-20-198.ec2.internal   <none>           <none>
frontend-79ccb5bc9f-ndsww   1/1     Running   0          131m   192.168.13.22   ip-192-168-20-198.ec2.internal   <none>           <none>
```

Describe ingress from my demo-joao:

```
joaoribeiro@JOCAPC:~/multi-tier-ingress-eks-lab$ kubectl describe ingress -n demo-joao
Name:             app-ingress
Labels:           <none>
Namespace:        demo-joao
Address:          adca49bf6fa4542a5a129dbe476c5321-e63121efeb8e4144.elb.us-east-1.amazonaws.com
Ingress Class:    nginx
Default backend:  <default>
Rules:
  Host                    Path  Backends
  ----                    ----  --------
  app.joao.diogohack.shop
                          /api   backend:3000 (192.168.2.132:3000,192.168.36.63:3000)
                          /      frontend:80 (192.168.13.22:80,192.168.31.107:80)
Annotations:              <none>
Events:                   <none>
```

Logs from my Backend app:

```
joaoribeiro@JOCAPC:~/multi-tier-ingress-eks-lab$ kubectl logs -n demo-joao -l app=backend --
tail=100

> backend@1.0.0 start
> node server.js

[backend] listening on http://0.0.0.0:3000

> backend@1.0.0 start
> node server.js

[backend] listening on http://0.0.0.0:3000
```

Logs from my frontend app

```
joaoribeiro@JOCAPC:~/multi-tier-ingress-eks-lab$ kubectl logs -n demo-joao -l app=frontend -
-tail=100
192.168.20.198 - - [06/May/2026:12:57:33 +0000] "GET / HTTP/1.1" 200 788 "-" "kube-probe/1.32" "-"
192.168.20.198 - - [06/May/2026:12:57:35 +0000] "GET / HTTP/1.1" 200 788 "-" "kube-probe/1.32" "-"
192.168.20.198 - - [06/May/2026:12:57:43 +0000] "GET / HTTP/1.1" 200 788 "-" "kube-probe/1.32" "-"
192.168.20.198 - - [06/May/2026:12:57:45 +0000] "GET / HTTP/1.1" 200 788 "-" "kube-probe/1.32" "-"
192.168.20.198 - - [06/May/2026:12:57:53 +0000] "GET / HTTP/1.1" 200 788 "-" "kube-probe/1.32" "-"
192.168.20.198 - - [06/May/2026:12:57:55 +0000] "GET / HTTP/1.1" 200 788 "-" "kube-probe/1.32" "-"
192.168.20.198 - - [06/May/2026:12:58:03 +0000] "GET / HTTP/1.1" 200 788 "-" "kube-probe/1.32" "-"
192.168.20.198 - - [06/May/2026:12:58:05 +0000] "GET / HTTP/1.1" 200 788 "-" "kube-probe/1.32" "-"
192.168.20.198 - - [06/May/2026:12:58:13 +0000] "GET / HTTP/1.1" 200 788 "-" "kube-probe/1.32" "-"
192.168.20.198 - - [06/May/2026:12:58:15 +0000] "GET / HTTP/1.1" 200 788 "-" "kube-probe/1.32" "-"
192.168.20.198 - - [06/May/2026:12:58:23 +0000] "GET / HTTP/1.1" 200 788 "-" "kube-probe/1.32" "-"
192.168.20.198 - - [06/May/2026:12:58:25 +0000] "GET / HTTP/1.1" 200 788 "-" "kube-probe/1.32" "-"
192.168.20.198 - - [06/May/2026:12:58:33 +0000] "GET / HTTP/1.1" 200 788 "-" "kube-probe/1.32" "-"
192.168.20.198 - - [06/May/2026:12:58:35 +0000] "GET / HTTP/1.1" 200 788 "-" "kube-probe/1.32" "-"
192.168.20.198 - - [06/May/2026:12:58:43 +0000] "GET / HTTP/1.1" 200 788 "-" "kube-probe/1.32" "-"
```

changing the diogohack.shop to ironlabs.online:

```yaml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: app-ingress
spec:
  ingressClassName: nginx
  rules:
    - host: app.joao.ironlabs.online
```

```
joaoribeiro@JOCAPC:~/multi-tier-ingress-eks-lab/k8s/k8s$ export HOST=app.joao.ironlabs.onlin
e
```

```
joaoribeiro@JOCAPC:~/multi-tier-ingress-eks-lab/k8s/k8s$ echo $HOST
app.joao.ironlabs.online
```

verify dns

```
dig +short $HOST
3.232.92.94
54.80.82.162
54.156.31.243
```

browser connection

Host = Pong

```
joaoribeiro@JOCAPC:~/multi-tier-ingress-eks-lab/k8s/k8s$ curl -s http://$HOST/api/ping
{"ok":true,"message":"pong","service":"backend","time":"2026-05-06T14:15:59.479Z","hostname":"backend-59947449d7-hc5xn
```